ORBIT SPACES OVER A LOCALLY FINITE TYPE PRE-RELATION DETECTED ON K-VALUED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This is the pointwise reincarnation of the orbit space definition under the locally finite type hypothesis.

Editorial note. The label lemma-orbit-space-locally-finite-type-over-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Definition definition-orbit-space (navigation stub): phi is an orbit space for R when it is R-invariant, separates R-orbits, and is surjective
- Chapter Definition definition-set-theoretically-invariant (navigation stub): set-theoretically R-invariant means U(k) to X(k) equalizes s and t for every algebraically closed k over B; separating orbits means equal values force the points into the same orbit
- Delivered card X2-alg-groupoids-quotients-geometric-orbits (Lemma lemma-geometric-orbits): under the locally finite type hypothesis, being in the same R-orbit on k-valued points agrees with equivalence under the relation generated by j(R(k))
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-finite-type: local finite type is preserved under composition
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-finite-type-surjective-geometric-points: a nonempty algebraic space locally of finite type over an algebraically closed field has a k-valued point

Source reference. Original reference 048V: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at 1429; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 048U: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 892–892. Prerequisite pointer invoked at 1431; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 048C: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 249–249. Prerequisite pointer invoked at 1401; it

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names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0487: sources/stacks-chow/tags.txt, source lines 5534–5534. Prerequisite pointer invoked at 1420; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0462: sources/stacks-chow/tags.txt, source lines 5457–5457. Prerequisite pointer invoked at 1417; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $B \to S$ as in Section 048C. Let $j = (t, s) : R \to U \times_B U$ be a pre-relation. Assume R, U are locally of finite type over B. Let $\phi : U \to X$ be an R-invariant morphism of algebraic spaces over B. Then ϕ is an orbit space for R if and only if the natural map*

$$U(k)/\big(\textit{equivalence relation generated by } j(R(k))\big) \longrightarrow X(k)$$

is bijective for all algebraically closed fields k over B.

Proof. Note that since U, R are locally of finite type over B all of the morphisms s, t, j, ϕ are locally of finite type, see Morphisms of Spaces, Lemma 0462. We will also use without further mention Morphisms of Spaces, Lemma 0487. Assume ϕ is an orbit space. Let k be any algebraically closed field over B. Let $\overline{x} \in X(k)$. Consider $U \times_{\phi, X, \overline{x}} \mathrm{Spec}(k)$. This is a nonempty algebraic space which is locally of finite type over k. Hence it has a k-valued point. This shows the displayed map of the lemma is surjective. Suppose that $\overline{u}, \overline{u}' \in U(k)$ map to the same element of $X(k)$. By Definition 048V this means that $\overline{u}, \overline{u}'$ are in the same R-orbit. By Lemma 048U this means that they are equivalent under the equivalence relation generated by $j(R(k))$. Thus the displayed morphism is injective.

Conversely, assume the displayed map is bijective for all algebraically closed fields k over B. This condition clearly implies that ϕ is surjective. We have already assumed that ϕ is R-invariant. Finally, the injectivity of all the displayed maps implies that ϕ separates orbits. Hence ϕ is an orbit space. □